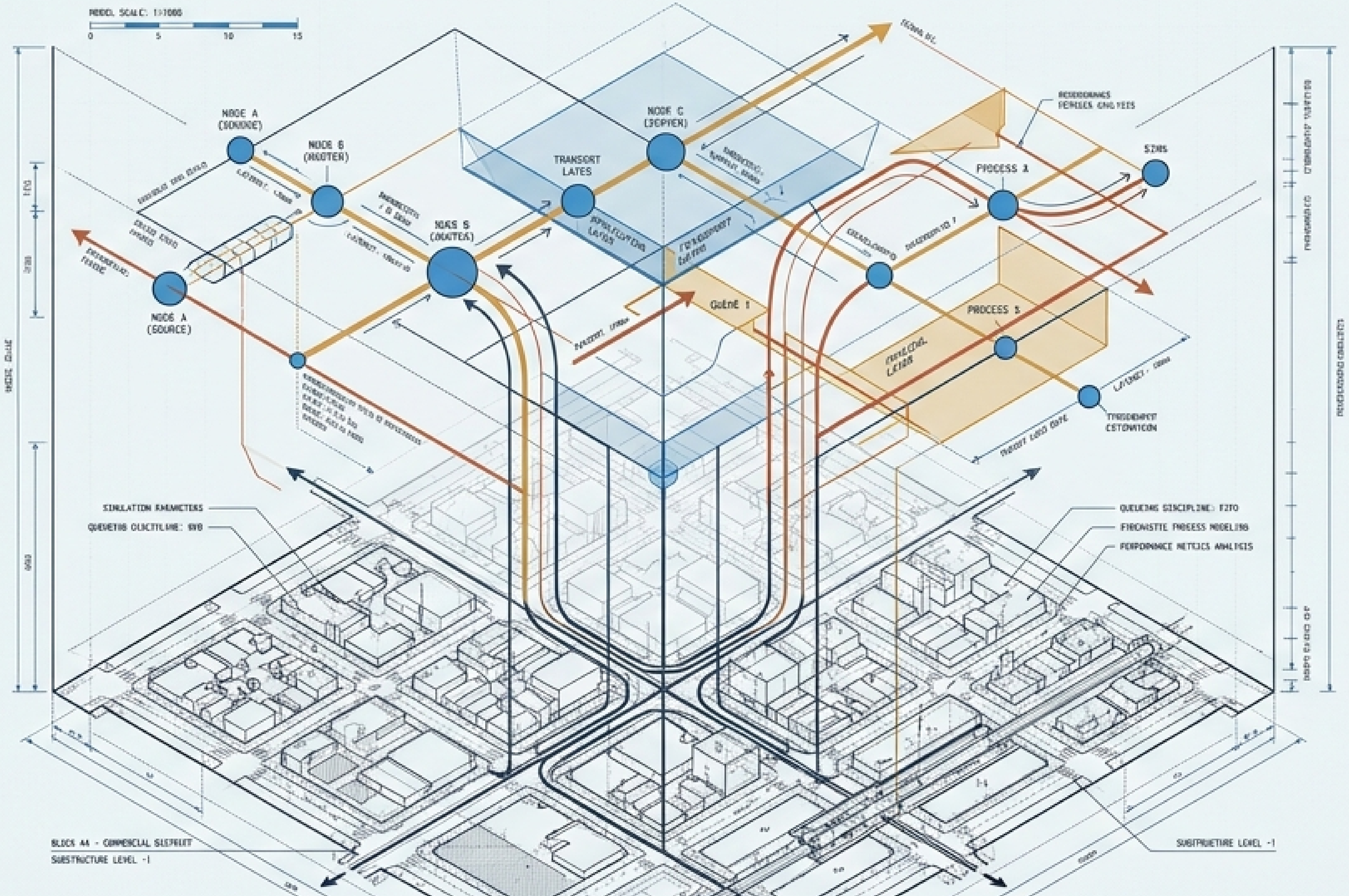
The Blueprint of Connectivity

Principles of Network Modeling & Performance Evaluation

A visual primer on queueing theory, stochastic processes, and architectural trade-offs.



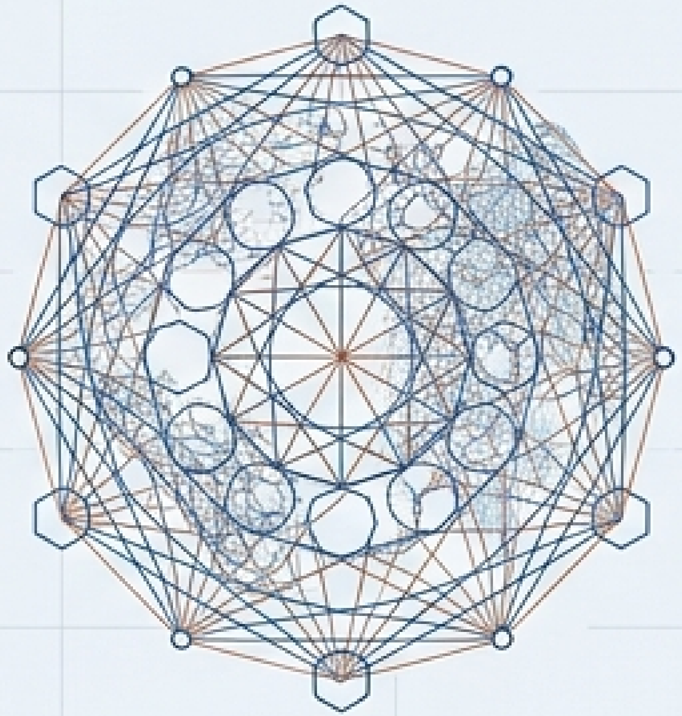
WHY WE BUILD NETWORK MODELS

PREDICT
$\lambda \uparrow \Rightarrow$ queue length $\uparrow \Rightarrow$ delay $\uparrow$

DESIGN
Compare 1 Gbit/s vs. 10 Gbit/s or routing algorithms before deployment.

OPTIMIZE
Minimize delay, loss, and energy while maximizing throughput and capacity.

REDUCE COST
Test thousands of scenarios without physical infrastructure.



THE MODELING ZONE

LOW COMPLEXITY

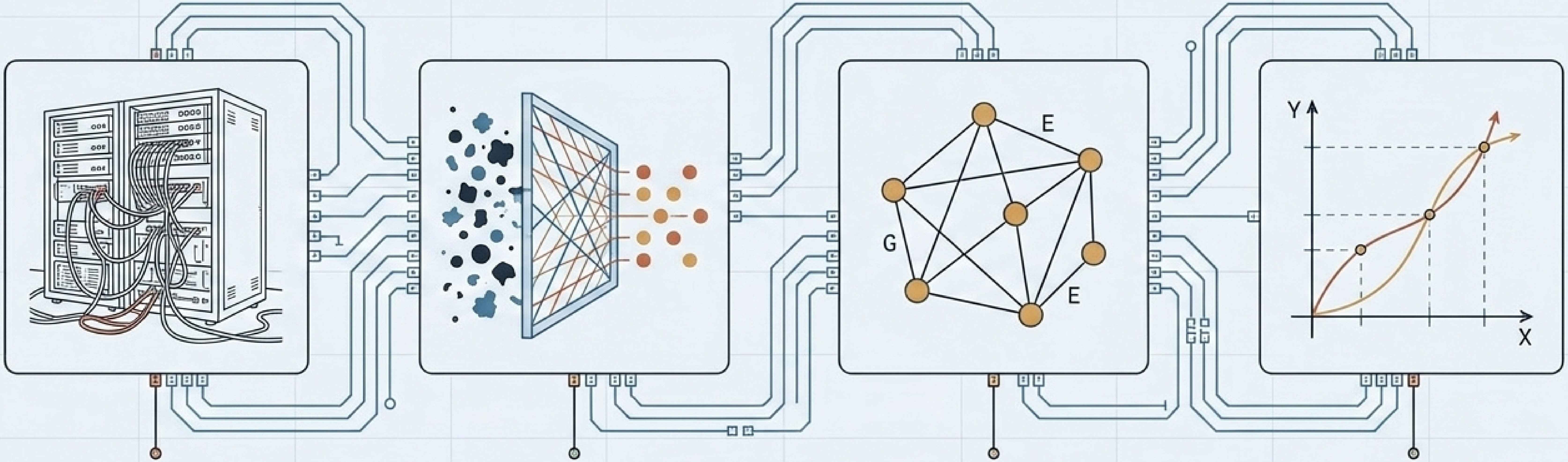
Easy to test physically.

Complexity Scale

HIGH COMPLEXITY

Impossible to test by trial and error.

THE ABSTRACTION PIPELINE



REAL NETWORK

The physical communication system.

ABSTRACTION

Reducing complexity while preserving properties that significantly influence performance.

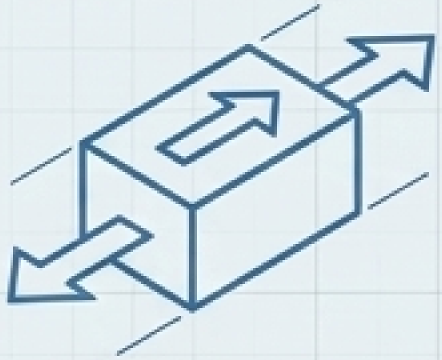
MATHEMATICAL MODEL

Simplified computational representation retaining relevant characteristics.

PERFORMANCE ANALYSIS

Extracting answers to capacity and behavioral questions.

ANATOMY OF A NETWORK MODEL



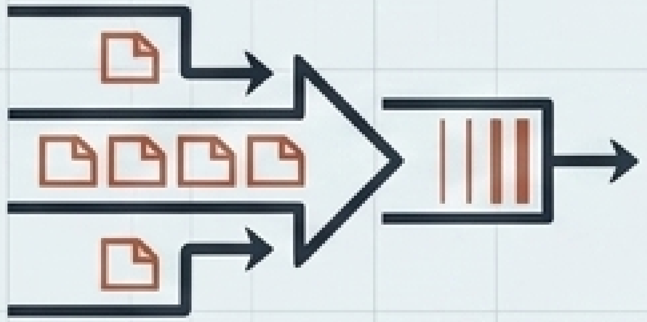
ENTITIES (WHO)

Objects participating in communication
(packets, flows, users, mobile terminals, routers).

RESOURCES (WHAT)

Elements used to provide services (transmission
links, buffers, radio channels, spectrum).

A resource becomes a bottleneck
when demand exceeds capacity.



WORKLOAD (HOW MUCH)

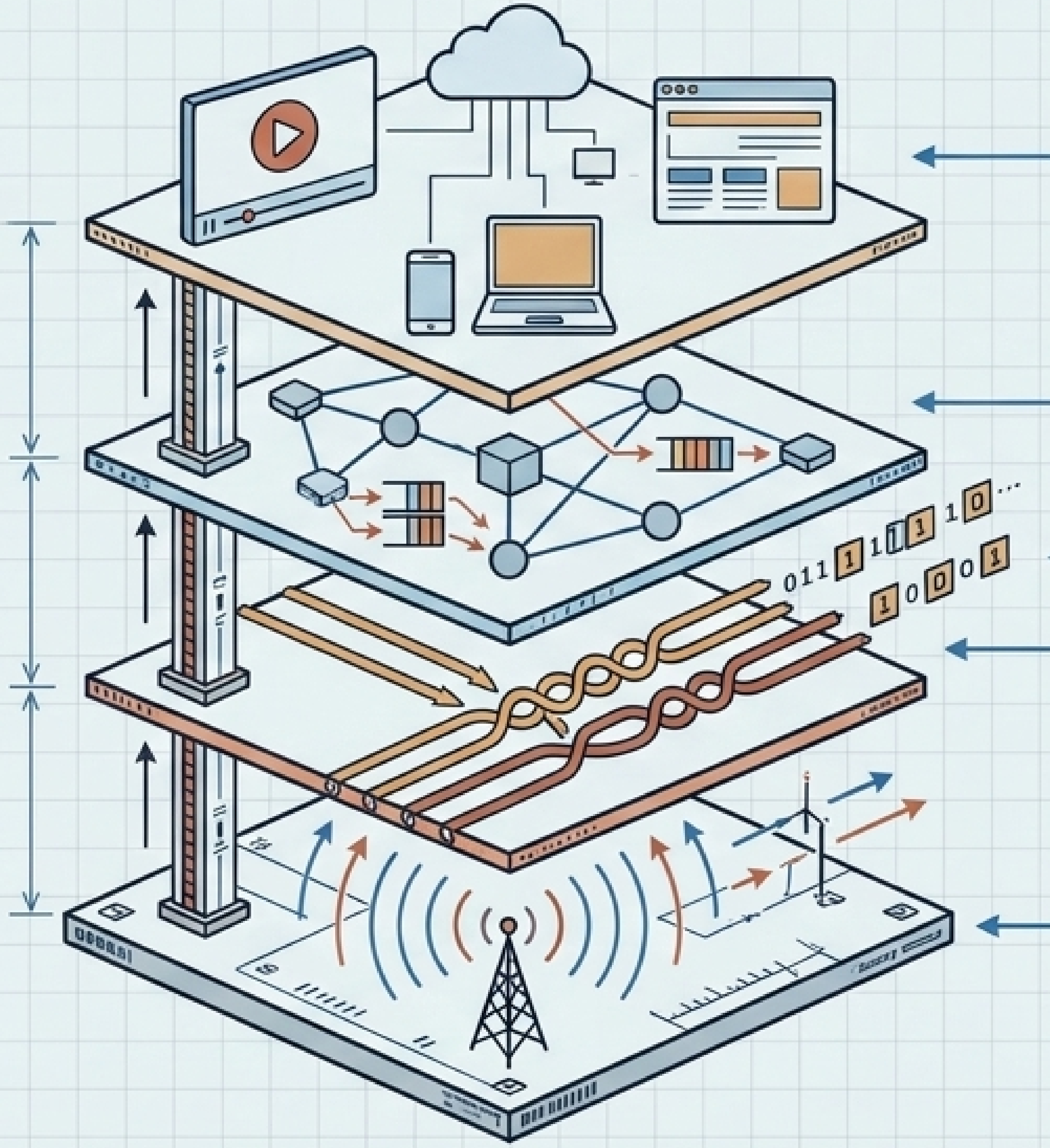
The traffic imposed on the network.
Defined by arrival rate λ = packets/time,
packet size, and burstiness.

PERFORMANCE MEASURES (HOW WELL)

Metrics quantifying system behavior
(throughput, delay, jitter, loss, utilization).

LEVELS OF ABSTRACTION

A good model is not necessarily the most detailed model. It contains sufficient detail to answer the specific question being studied.



Application Layer (Streaming / Web)

Models user and software behavior.

Network Layer (Queues / Routing)

Models topology, congestion, and packet forwarding.

Link Layer (Bits / Errors)

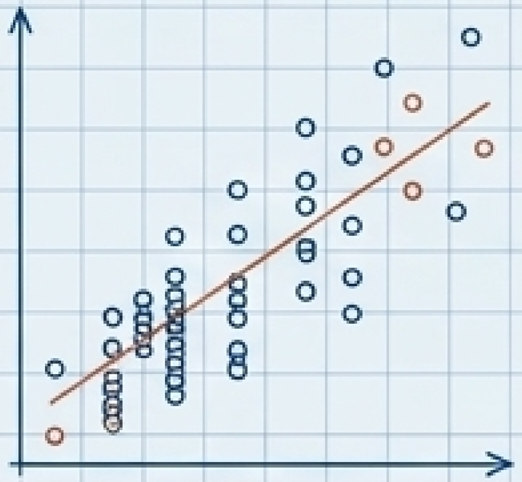
Models bandwidth, data rate, and bit error rate.

Physical Layer (Waves / Power)

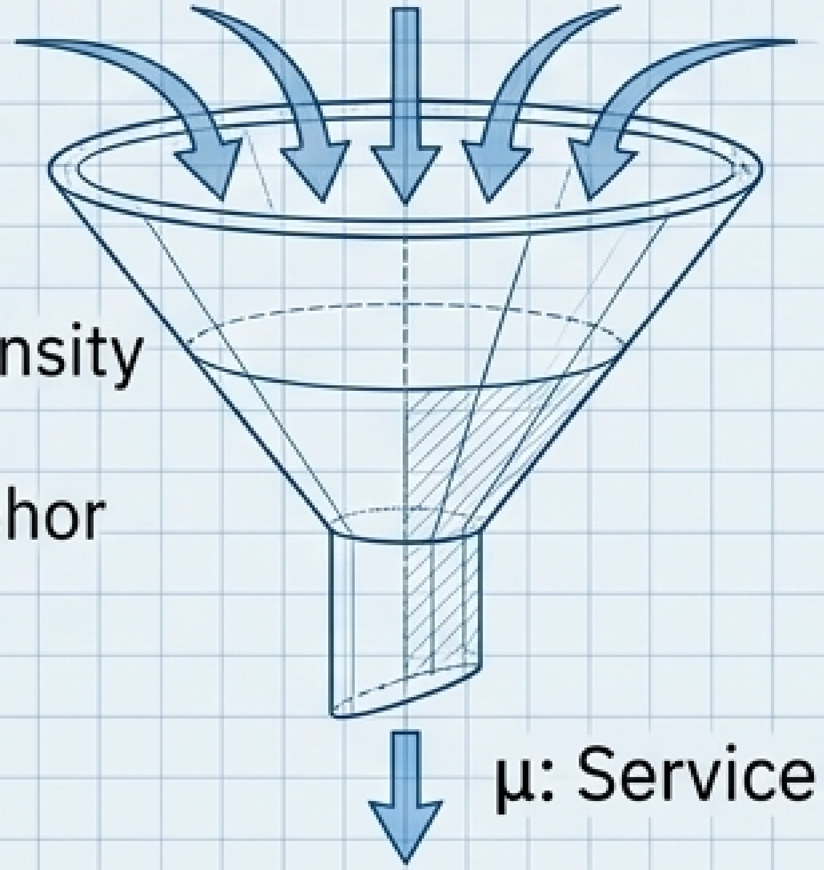
$$\boxed{P_r = P_t\, G_t\, G_r\, L}$$

Models signal power, noise, and interference.

The Nature of Traffic: Deterministic vs. Stochastic

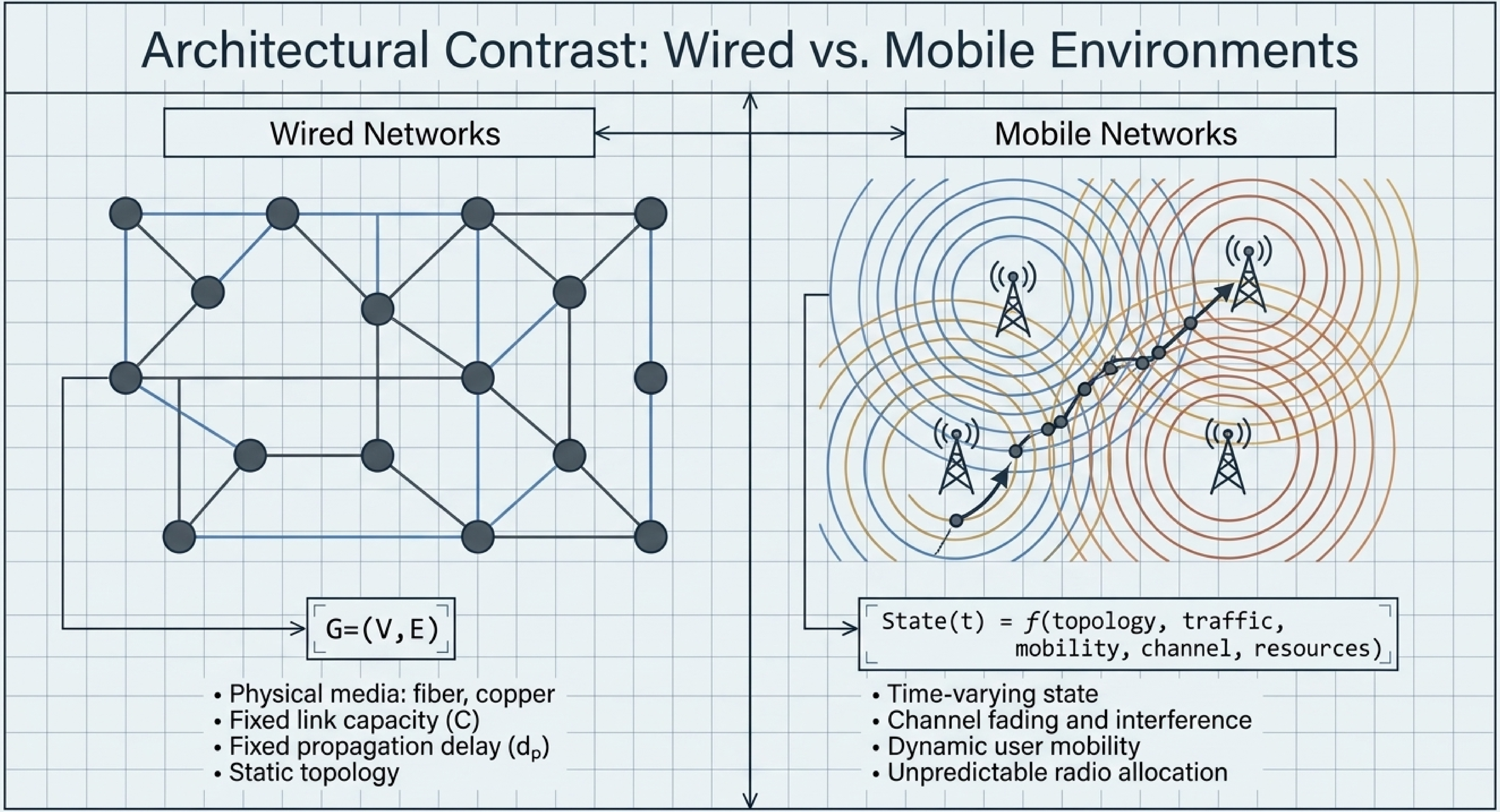
Deterministic	Stochastic
 Perfectly predictable arrival times. Ta = 10 ms	 Represents real-world uncertainty. Arrivals A(t) ~ Poisson Service S ~ Exponential(μ)

“Traffic Intensity Funnel” Hero Metaphor



Traffic Intensity: $\rho = \dfrac{\lambda}{\mu}$

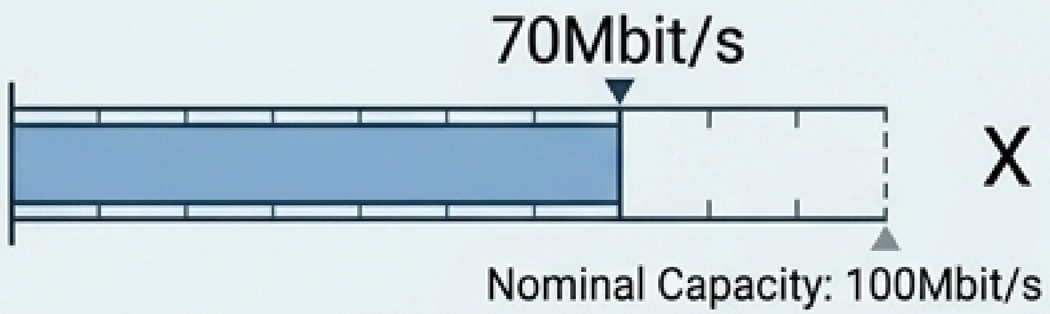
> ⚠ **Critical Rule:** Stable operation requires $\rho < 1$. If $\rho \geq 1$, an unlimited queue will not have a stable long-term average size.


Architectural Contrast: Wired vs. Mobile Environments
Wired Networks
Mobile Networks
G=(V,E)
State(t) = f(topology, traffic, mobility, channel, resources)
• Physical media: fiber, copper
• Fixed link capacity (C)
• Fixed propagation delay (d_p)
• Static topology
• Time-varying state
• Channel fading and interference
• Dynamic user mobility
• Unpredictable radio allocation

The QoS Metric Dashboard

There is no single universal metric. QoS depends entirely on the application.

Throughput



$$X = \frac{\text{useful data}}{t}$$

Throughput ≠ Nominal Capacity. A 100Mbit/s link may only yield 70Mbit/s due to overhead.

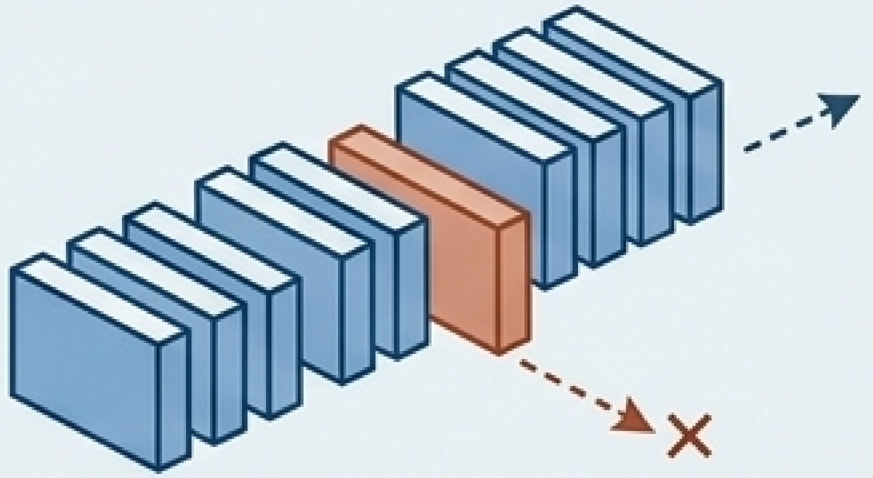
Utilization

$$U = \frac{R}{C}$$

(Carried Traffic / Capacity)

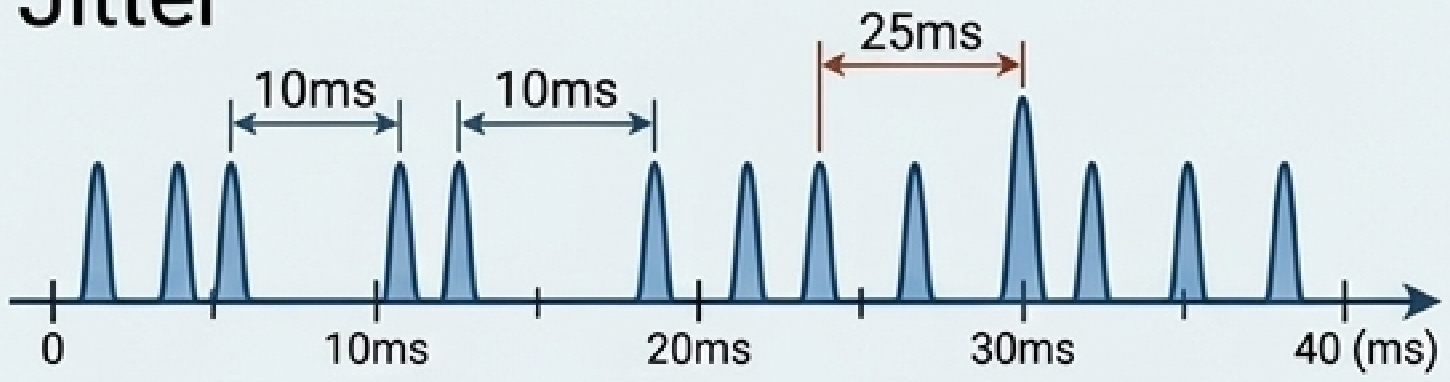
High utilization is efficient, but excessive spikes delay.

Packet Loss Ratio



$$P_{loss} = \frac{N_{lost}}{N_{sent}}$$

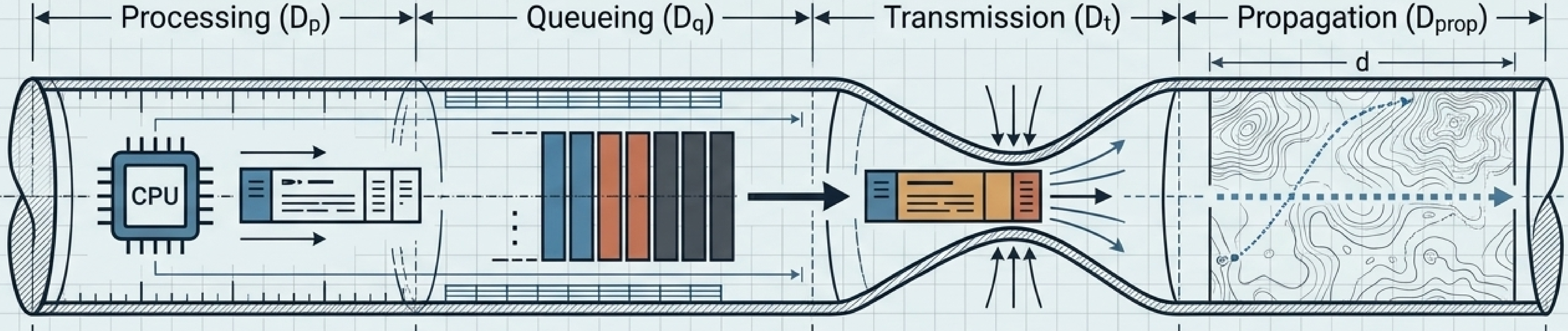
Jitter



Delay variation. A sudden spike from 10ms to 25ms severely disrupts real-time apps.

The Anatomy of Delay

$$D_{total} = D_p + D_q + D_t + D_{prop}$$



Processing (D_p)	Queueing (D_q)	Transmission (D_t)	Propagation (D_{prop})
Time required to process a packet.	Time spent waiting in buffers.	$D_t = L/R$ (Packet bits L / Transmission rate R)	$D_{prop} = d/v$ (Distance d / Speed v)

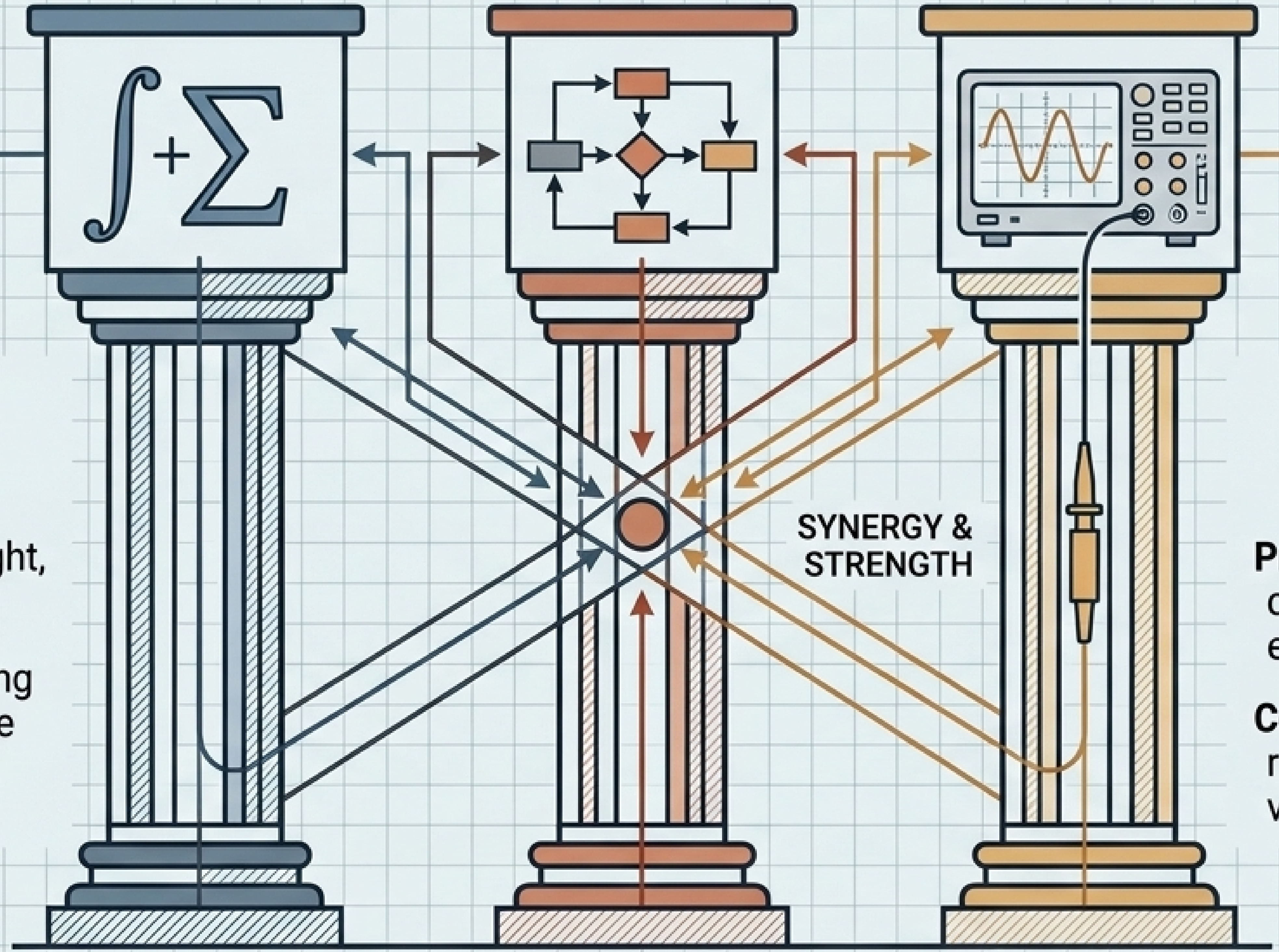
The Evaluation Triad

The strongest studies combine Analytical + Simulation + Measurement.

Analytical Modeling

Pros: Mathematical insight, exact solutions, speed.

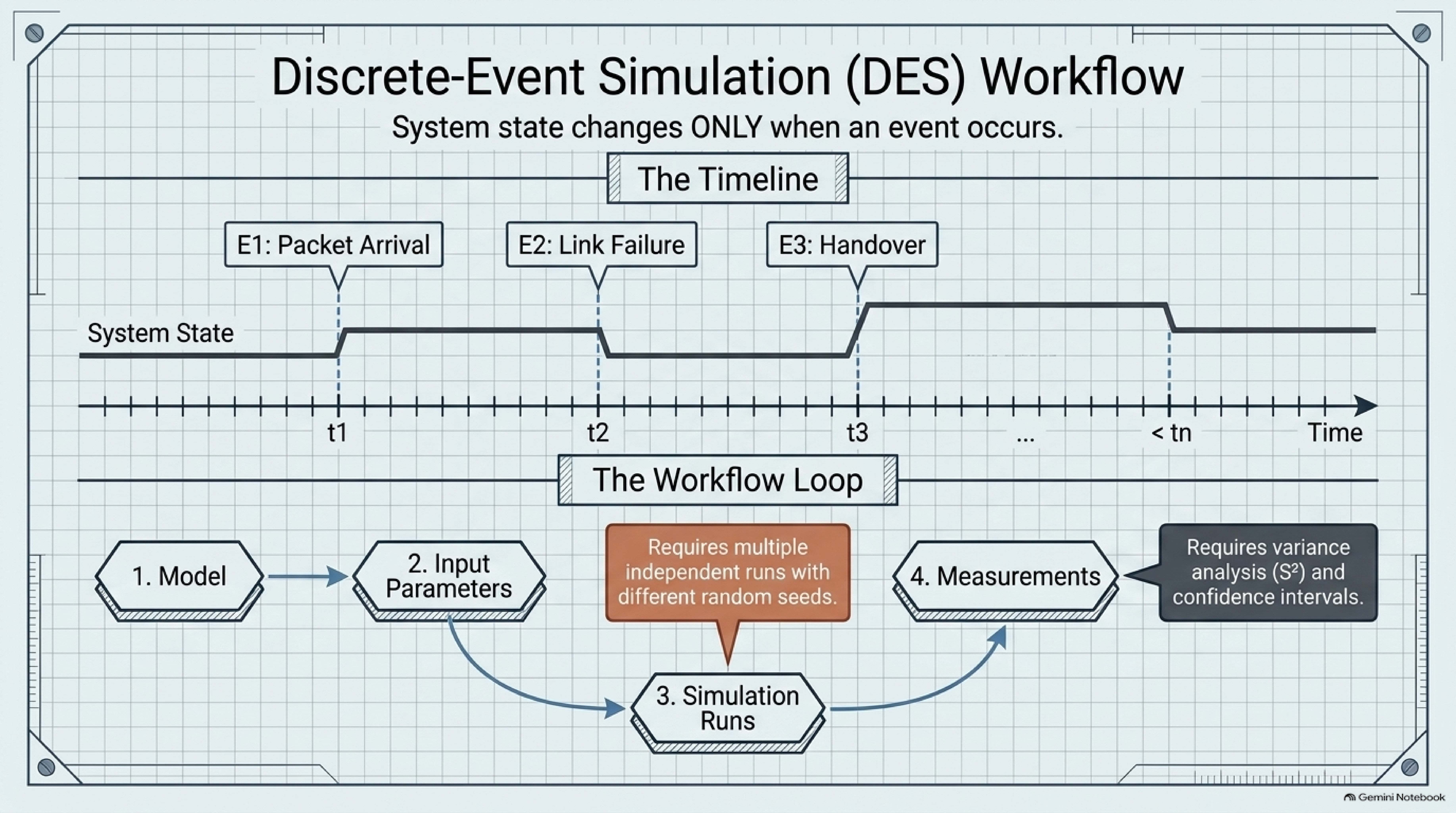
Cons: Requires simplifying assumptions, intractable for extreme complexity.



Experimental Measurement

Pros: Real-world accuracy, captures hidden physical effects.

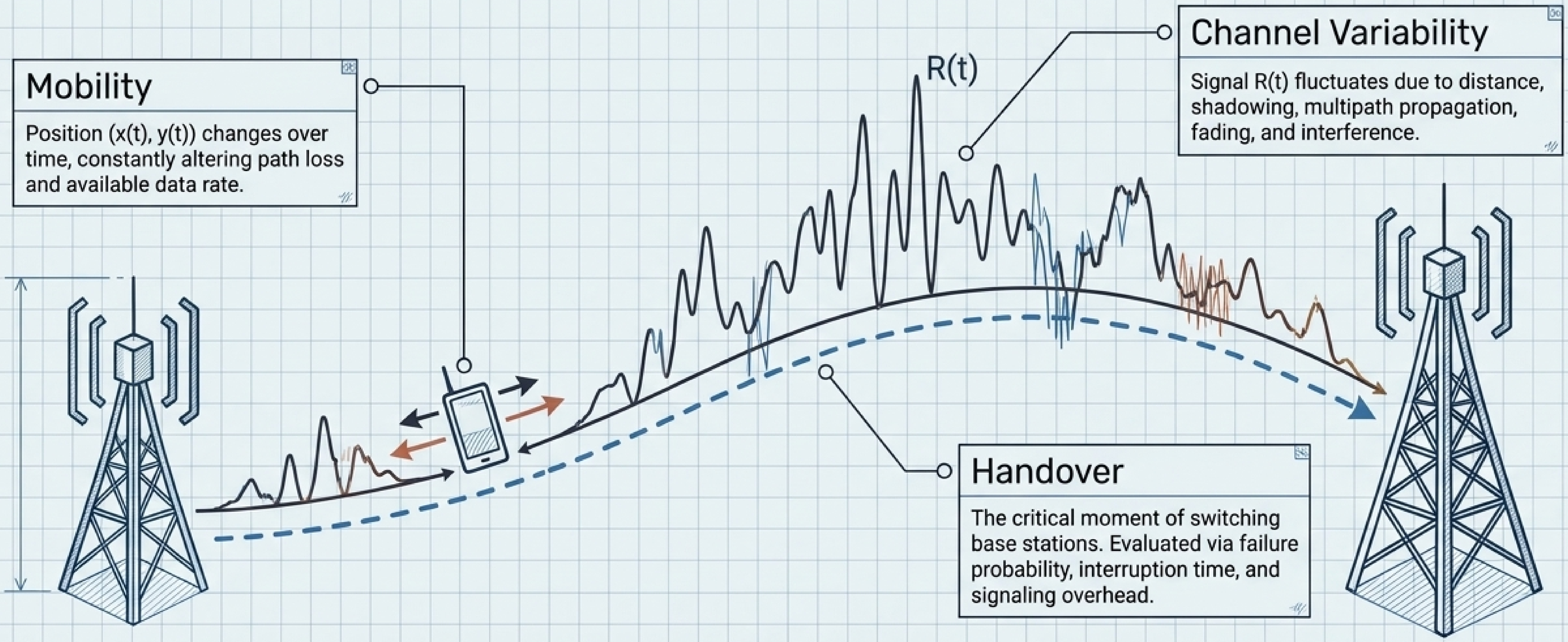
Cons: Expensive, limited reproducibility, poor variable control.



Discrete-Event Simulation (DES) Workflow
System state changes ONLY when an event occurs.
The Timeline
E1: Packet Arrival
E2: Link Failure
E3: Handover
System State
t1
t2
t3
...
< tn
Time
The Workflow Loop
1. Model
2. Input Parameters
Requires multiple independent runs with different random seeds.
3. Simulation Runs
4. Measurements
Requires variance analysis (S²) and confidence intervals.
Gemini Notebook

The Mobile Complexity Multiplier

$$C = f(\text{spectrum}, \text{bandwidth}, \text{SINR}, \text{reuse}, \text{scheduler}, \text{traffic})$$



Verification vs. Validation

Verification ≠ Validation (A perfectly implemented simulation can still be based on a poor model).

Verification

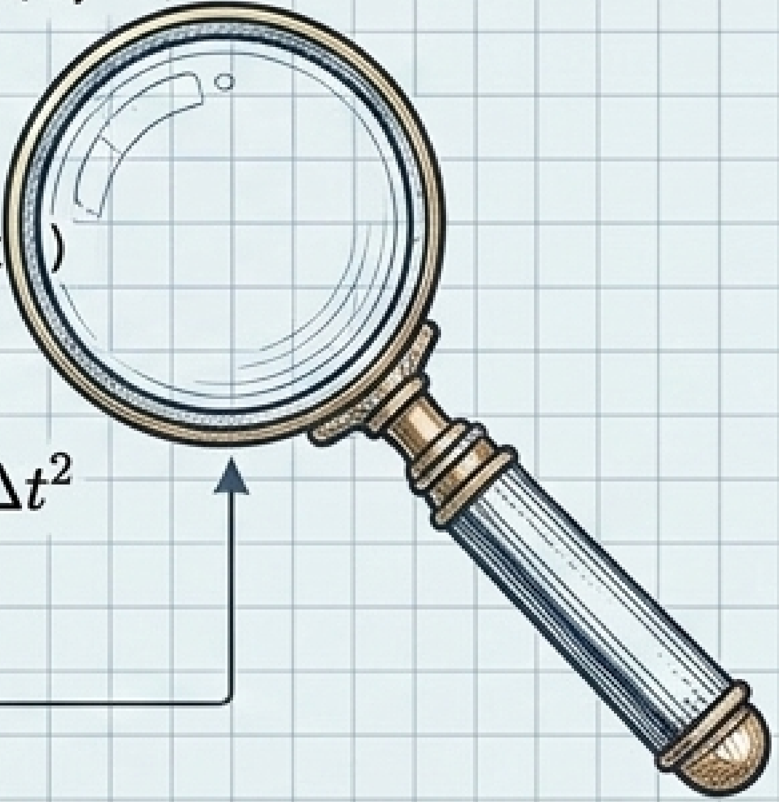
Did we build the model right?

```
def calculate_position(t, v):
    return v * t

if state == 'waiting':
    process_next_event()
```

$$\Delta x = v_0 * \Delta t \boxed{+} \frac{1}{2} * a * \Delta t^2$$

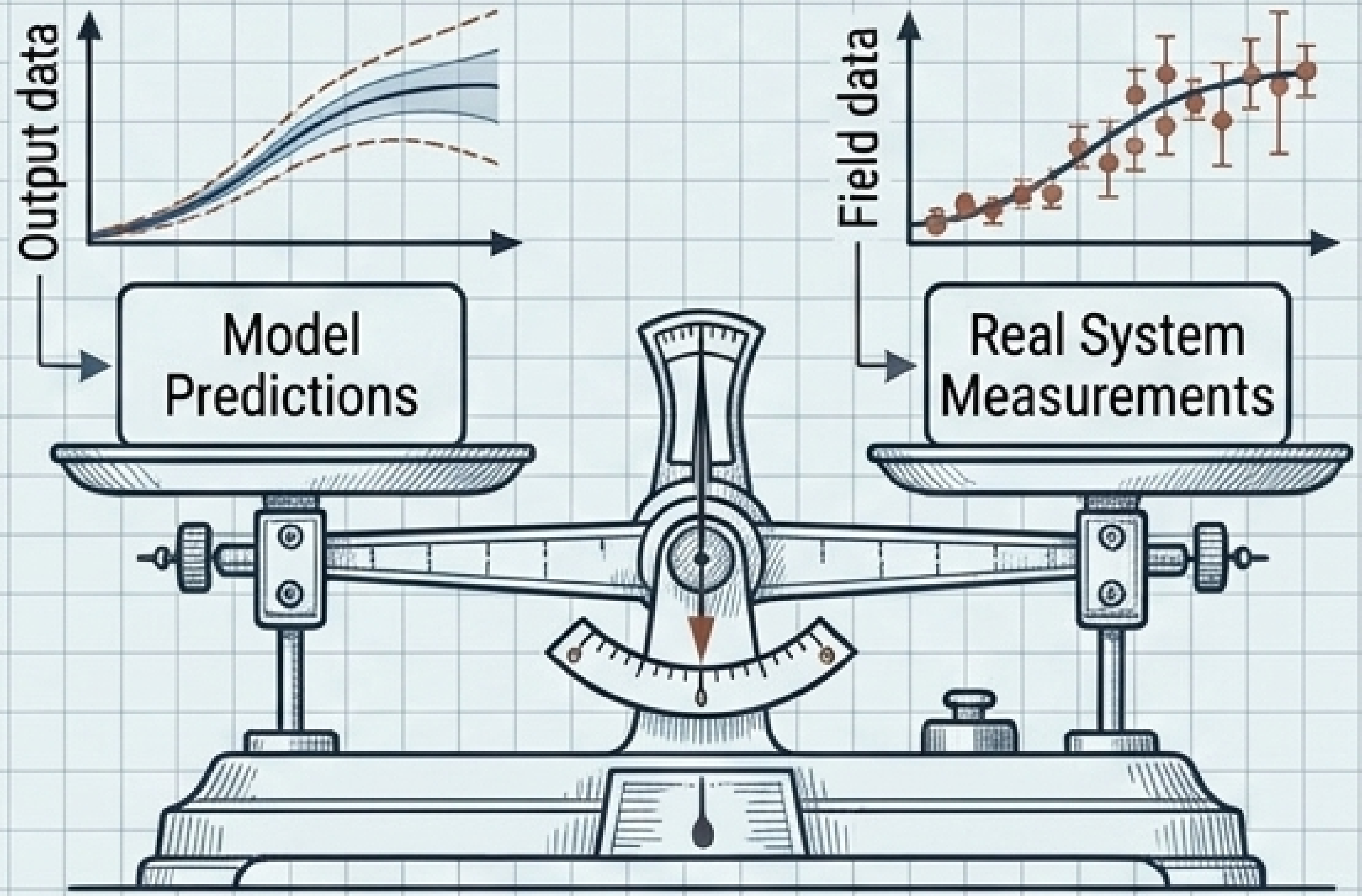
$$f(t) = \int_0^t e^{-\tau} d\tau$$



Checking the correctness of the implementation. Ensuring there are no bugs in the simulation logic or software.

Validation

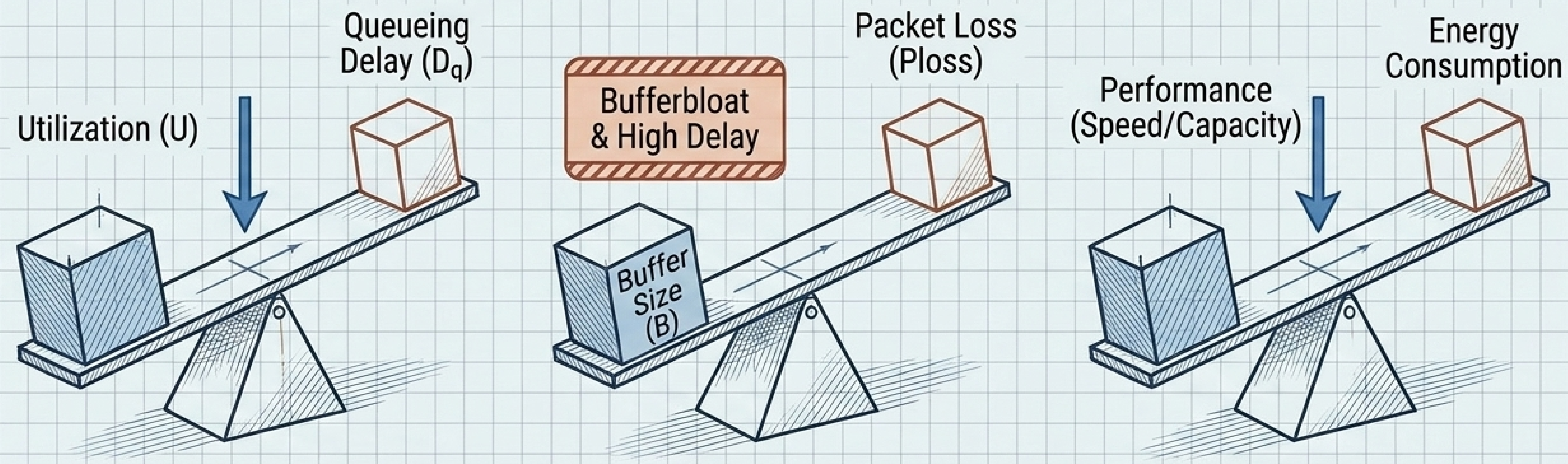
Did we build the right model?



Determining if the model adequately represents reality for its intended purpose.

The Immutable Trade-Offs

Network engineering is fundamentally about compromise. You cannot optimize every metric simultaneously.



Approaching saturation yields high efficiency but spikes delay.

Large buffers stop drops but trap packets in queue.

Higher mobile throughput requires heavier power drains for transmission and signaling.

Synthesis: The Methodology

1. Define Objective

2. Define System
(topology, resources)

3. Define Workload
(λ, traffic patterns)

4. Select Metrics
(X, D, Ploss, U)

5. Choose Method
(Analytical, Sim, Measure)

6. Run Experiments
(Vary λ, N, C, B)

7. Analyze Results
(Find bottlenecks)

8. Validate
Conclusions

Mentor's Perspective

Never memorize a formula without asking:
What physical phenomenon does this describe?
When does it stop being a good approximation?

The ultimate goal is not to calculate a metric, but to answer: Why does the network behave this way, and how can we improve it?